Product Requirements Document (PRD)

SHAKTI AI – AI-Based Tender Evaluation & Eligibility Analysis Platform

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1. Product Overview

SHAKTI AI is an intelligent procurement platform designed to automate the evaluation of government tenders, specifically for organizations like CRPF. The system leverages AI, OCR, and rule-based engines to extract eligibility criteria from tender documents and evaluate bidder submissions with high accuracy and full explainability.

The platform addresses inefficiencies in manual tender evaluation, reducing time, eliminating inconsistencies, and ensuring transparency in procurement decisions.

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2. Problem Statement

Government tender evaluation today faces multiple challenges:

- Manual review of large, complex documents
- Inconsistent decision-making across evaluators
- Difficulty in handling scanned and unstructured documents
- Lack of transparency and auditability
- High time consumption (days per tender)

There is a need for an automated, explainable, and auditable AI-driven solution.

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3. Objectives

- Automate extraction of eligibility criteria from tender documents
- Analyze bidder submissions across multiple document formats
- Provide criterion-level eligibility decisions (Eligible / Not Eligible / Review)
- Ensure full explainability and audit trail
- Reduce evaluation time by 70–80%
- Enable human-in-the-loop decision making

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4. Target Users

Primary Users:

- Government Procurement Officers (CRPF Admin Panel)

Secondary Users:

- Private Bidders / Vendors (Bidder Panel)

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5. Key Features

5.1 Tender Understanding (AI Criteria Extraction)

- Extract eligibility criteria using AI (LLM + NLP)
- Categorize into:
    - Financial
    - Technical
    - Compliance
    - Experience

- Identify:
    - Mandatory vs Optional
- Output structured JSON

5.2 Bidder Document Processing

- Support:
    - PDFs
    - Scanned documents
    - Images
    - Word files
- Use OCR (Tesseract)
- Extract:
    - Financial values
    - Certificates
    - Experience data

5.3 AI Evaluation Engine

- Match bidder data against criteria
- Generate decision:
    - Eligible
    - Not Eligible
    - Needs Review

- Handle:
  - Partial data
  - Missing data
  - Ambiguity

5.4 Explainable AI Decisions

Each decision must include:

- Criterion checked
- Extracted value
- Source document
- Reason for decision
- Confidence score

5.5 Human-in-the-Loop System

- Flag low-confidence cases
- Manual review dashboard
- Actions:
  - Approve
  - Reject
  - Request Clarification

5.6 Fraud Detection (Advanced Feature)

- Detect:
  - Duplicate GST
  - Same address/phone
  - Suspicious patterns
- Show risk score

5.7 Dashboard & Reporting

- Admin dashboard:
  - Total bidders
  - Eligible / Rejected / Review
- Export reports:
  - PDF
  - Excel
- Full audit trail

5.8 Bidder Dashboard

- View available tenders
- Upload documents
- Track evaluation status
- Respond to clarifications
- Download reports

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6. System Architecture

Frontend

- React.js (Dashboard UI)
- Tailwind CSS (Styling)

Backend

- FastAPI (Python)
- REST APIs

Database

- PostgreSQL (Neon Cloud)

AI Layer

- LLM (Gemini / OpenAI)
- NLP for classification

OCR Engine

- Tesseract OCR
- OpenCV preprocessing

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7. Workflow

1. Upload Tender Document
2. OCR + Text Extraction
3. AI extracts criteria (JSON)
4. Upload Bidder Documents
5. Extract bidder data

6. Match with criteria
7. Generate decisions
8. Show results with explanations
9. Flag uncertain cases for review

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8. Non-Functional Requirements

- High accuracy (>90%)
- Fast processing (<1 min per document)
- Secure (JWT authentication)
- Scalable architecture
- Full audit logging
- Reliable OCR handling

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9. Risks & Challenges

Risk	**Solution**
Poor OCR quality	Image preprocessing + manual review
Ambiguous language	Confidence scoring + human review
API latency	Async processing + caching
Data inconsistency	AI normalization

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10. Success Metrics

- 80% reduction in evaluation time

- 90% accuracy in eligibility decisions
- 100% explainability
- Reduced manual workload
- Improved transparency

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11. Future Enhancements

- Multilingual tender support
- Advanced fraud detection (graph AI)
- Integration with government e-procurement systems
- AI-based bidder ranking

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12. Conclusion

SHAKTI AI transforms traditional tender evaluation into a fast, intelligent, transparent, and auditable process. By combining AI, OCR, and explainable decision-making, it empowers government agencies to make consistent and reliable procurement decisions.
